Voice-Based Email for Blind Users — Prior Work Comparison

How this senior project differs from existing systems and research (strengths, weaknesses, and my contribution)

Project / System	Approach	Strengths	Weaknesses	How My Project Differs
Voice-Based Email System (2016, IJSRSET)	Keyword-triggered STT/TTS menu	Early proof it's technically feasible; simple, low-resource	Rigid command grammar; user must memorize exact keywords; no context/memory between commands	Replaces fixed keywords with LLM-parsed natural sentences — no memorization needed
Voice Based Mail System for Visually Impaired (2022, IJERT)	Menu-driven speech app, third-party-assisted fallback	Documents real pain points of blind users relying on sighted helpers	Still menu/keyword-based; assumes human backup when voice fails; no smart error recovery	Handles ambiguity itself (asks a clarifying question) instead of falling back to a human helper
voice-based-email-for-blind (GitHub, hacky1997)	Flask + Web Speech API + Gmail OAuth, PWA	Good technical reference architecture; real working IMAP/SMTP integration	No NLP/intent layer — commands are likely still fixed phrases; no evaluation with real users	Adds an LLM intent layer on top of a similar architecture, plus user testing they didn't do
TetraMail (2020, empirical study, 38 blind participants)	Screen-reader + touch hybrid, not pure voice	Only project here with a rigorous usability study; real accuracy/task-completion data	Touch-dependent (not hands-free); pre-LLM, so no natural language; screen-reader-based, not standalone voice control	Voice-first (hands-free) and natural-language driven; reuses their evaluation methodology for my own study
Speaking Email (commercial app, iOS/Android)	TTS content extraction + voice commands + touch gestures	Polished, shipped, well-reviewed; smart content extraction (strips signatures/threads)	Closed-source, can't build on or cite as technical work; command set still fairly fixed, not conversational	Matches its content-extraction quality but adds open, LLM-based conversational commands, documented and publishable
AURA (2026, Scientific Reports)	LLM-guided voice interaction + real-time behavior personalization	Recent, rigorous, shows LLM+accessibility is a live research area; adapts to individual user behavior	General-purpose assistant, not built for a specific task like email; no human-subject evaluation done (acknowledged limitation)	Takes their “LLM voice interaction for blind users” concept but applies it to a concrete task (email) with real user testing
“I'm Blind, ChatGPT” (2025, ACM CUI)	Interaction study of blind users with LLM conversational AI (a study, not a built app)	Documents real breakdown/repair patterns: delays, unclear system state, audio channel collisions with screen readers	Not a system at all — pure interaction research, no app to compare against	Designs directly around their documented failure modes (e.g., prevents TTS from colliding with the phone's native screen reader)
Insight (mobile accessibility, arXiv)	LLM analyzes screen content, gives voice summaries, acts on user's behalf	Natural language commands, screen-content awareness	General mobile navigation/typing assistant, not email-specific; no dedicated email workflow (compose/reply/search/delete)	Goes deep on one task (email) rather than broad/shallow across the whole OS — allows proper confirmation/undo flows specific to email actions

Project Thesis

Prior voice-email systems (2016–2022) use rigid keyword commands with no natural language understanding. Recent LLM-based accessibility research (2025–2026) explores natural language interaction but focuses on general screen navigation or scene description, not email specifically. No existing system combines LLM-based natural language command parsing with a dedicated, evaluated email workflow for blind users — that gap is what this project addresses.